

CURVATURE-RATIO LOCK-IN FOR BARYON ASYMMETRY: DEMONSTRATED ON THE HQIV DISCRETE SPINE (LEAN FORMALISATION)

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ABSTRACT. This note—the seventh HQIV preprint—is a capability demonstration on the shared discrete curvature-ratio spine. Baryon-asymmetry bookkeeping is already a theorem-level consequence of the same lock-in calibration, monotonicity results, and triality scaffolds used for hadron structure, BBN, and other applications in the library. At lock-in shell $m = 4$, horizon ratios $\Omega_k(n; N) = I(n)/I(N)$ are proved monotone with $\Omega_k(M; M) = 1$, and the baryon readout $\eta(n; N)$ shares that normalisation; a single quarantined anchor η_{paper} supplies the comparison-layer scale. The primary artefacts are the Lean formalisation and the Python reproduction bundle (§11). Programme context, witness panels, and the proved/witnessed/deferred split are in §1; Lean module and identifier names are in Appendix A.

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as $\text{Fin } 4$, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation $(Q_L, u^c, d^c, L, e^c, \nu^c)$, the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite $\text{Fin } 3$ label sum of the same cancelled one-generation block, not a fitted multiplicity.
- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $G_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour commutator certificates are discharged on the octonion carrier (see the sector zeta and holonomy discharge bundle in the Lean catalog). ****Matrix Wilson layer:**** full four-edge $SO(8)$ generator transport and Hopf-shell T^4 torsion plaquettes with pinned non-abelian witness ([Hqiv/Physics/SO8PlaquetteHolonomy.lean](#), witness `so8PlaquetteHolonomyDischarged_holds`). ****Wilson-kinetic equivalence**** ([Hqiv/Physics/ActionHolonomyGlue.lean](#), witness `wilsonKineticPlaquetteEquivalence_discharged`). ****Partial / open:**** Haar measure on rotated charts ([Hqiv/Physics/NonAbelianHolonomyMeasureScaffold.](#)

[lean](#)); general flat HQVM kinetic boost and full strong-interacting discrete-action Poincaré ([Hqiv/Physics/DiscreteActionStrongPoincareBridge.lean](#), [Hqiv/Physics/DiscreteActionPoincareScaffold.lean](#)). Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart.

- **Rapidity phase and Lorentz invariance.** The null-lattice carrier `NullLatticeEvent` charts into 1+1 Minkowski space by the classical rapidity parametrization $\Lambda(\eta)(1, 1)$; rapidity addition on the carrier is equivariant under `boostMatrix11`, the Minkowski pairing is bilinearly invariant, and the polar-angle / `zetaHQIVTerm` phase is unchanged when ϕ transforms as a Lorentz scalar. The honest 3+1 extension preserves `minkowskiSq4` on the embedded $(t, x^1, 0, 0)$ plane.
- **Spatial rotations and directional readouts.** Orthogonal $O(3)$ rotations on `Fin3` spatial vectors preserve Euclidean inner products, $\|\mathbf{r} \times \mathbf{v}\|^2$, axis-projection fractions, and the spatial block of `minkowskiSq4` when time is fixed ([Hqiv/Geometry/SpatialRotationLorentzClosure.lean](#), witness `spatial_rotation_lorentz_closure_discharged`). Applied discharge: CMB sky rotation shifts the Stokes angle additively while preserving $\cos^2 + \sin^2 = 1$; O-Maxwell fluid sources rotate covariantly with gradients; coronal axis projections are joint rotation invariants. Full patch Lorentz on the flat chart is the rapidity bundle plus this spatial bundle (witness `full_lorentz_closure_discharged`). Continuum interacting QFT / Wightman theorems are not claimed.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and sector zeta bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

Projection geometry and the classical ζ dictionary (2026-06). HQIV uses *zeta* in two distinct, now formally separated roles:

- **Sector zeta on the patch carrier** (TUFT+SM synthesis): fibre-torsion subleading bodies, `zetaHQIVTerm` phase on Hopf shells, and holonomy discharge bundles on the octonion light-cone carrier—finite spectral readouts indexed by shell and Fano sector, not the Riemann function.
- **Classical $\zeta(s)$ as an $SO(4)$ projection readout** ([13]): the Mathlib Riemann zeta packaged as a *regional* closed form on $\mathbb{C}$ —Dirichlet/Euler for $\Re s > 1$, sin-cos- Γ assembly on the strip, Bernoulli slots—with the functional-equation pair $(\sigma, 1 - \sigma)$ interpreted as a 45° quaternionic projection. Machine-checked on this spine: harmonic divergence forces Δ and $\mathfrak{so}(4)$ closure; shell sum $= \zeta(s)$ for $\Re s > 1$; polar decoupling $n^{-s} = n^{-\sigma} \cdot e^{-it \log n}$ (modulus σ -only, phase t -only); prime-log Diophantine independence; Goldbach circle $pq + (q - N)^2 = N^2$ as additive curvature in the log channel; and a large family of RH-*equivalent* locators (twiddle, spectral weights, polar collapse, transformer adjoint law, S^7 torsion cancellation) with *zero slack*—not a discharge of RH or Goldbach parity.

Continuum heat-flow and de Bruijn–Newman notation in the same story spine are *comparison ports* (`dbnHeatFamily`, lattice dominance hypotheses); they are not load-bearing continuum extensions of the patch. Formal mathematical notes may include only the readout-dictionary paragraph (`readout_dictionary_messaging.tex`) rather than this full contract when the manuscript is projection geometry rather than patch phenomenology.

1. INTRODUCTION

1.1. Programme context and doorway. This note is a **Tier-2 checkpoint**, not the programme’s goal. The HQIV objective is a discrete curvature-ratio ledger on the null lattice—with sector projections to hadronic binding, electroweak synthesis, chemistry, compact objects, and cosmology—not a standalone theory of baryogenesis. Baryon-asymmetry bookkeeping is one cosmological *application*: it records how the same lock-in calibration, Ω_k monotonicity, and quarantined η_{paper} anchor answer a single ledger question at shell $m = 4$.

Geometry and ratios live in the core formal modules ([octonionic lightcone](#), [baryogenesis core](#), [baryogenesis witness](#); Appendix A); η bookkeeping is how that spine answers one comparison-layer question. Readers should treat this note as an on-ramp to the larger corpus, not as an isolated claim about observed η alone.

- **Plain-language tour:** [disregardflat.tech](#) (clickable module anchors).
- **Formal library:** [HQIV-LEAN](#) [14].
- **Reproduction bundle:** `scripts.zip` in this folder (§11).
- **Companion BBN stack:** [3].

1.2. Publication chain. Tier 0 fixes the framework [6, 8, 10]; Tier 1 supplies the variational and lattice spine [12, 9, 7, 11]; Tier 2 adds the published Lagrangian bridge [5]. The present note states the strongest *ratio-level* baryogenesis theorem currently supported: baryogenesis is not an independent knob—it is the same partial curvature imprint as the growth channel, triality CP-bias scaffolds, and the HQVM Friedmann compatibility package.

The same spine already shows machine-checked readouts across applied scales—nucleon binding [4], TUFT+SM synthesis [5], chemistry and phase geometry, compact-object witnesses, and the BBN integrator [3]. This note records how the baryon-asymmetry ratio slots into that spine at lock-in.

1.3. Formal results (summary). Discrete shell combinatorics define a cumulative curvature integral $I(n)$; horizon ratios $\Omega_k(n; N) = I(n)/I(N)$ are machine-checked with proved monotonicity at fixed N ; the lock-in shell $m_{\text{lockin}} = M = 4$ calibrates $\Omega_k = 1$ at the lock-in horizon; and $\eta(n; N)$ is *defined* to share that same normalisation—the convention the companion BBN note [3] takes at lock-in. Light-element abundances are *not* claimed here.

1.4. Quarantined anchor and reader contract. The observed-order scale η_{paper} appears in exactly one formal module ([paper \$\eta\$ anchor module](#)) and is wired only through the witness layer. Ratio laws and lock-in calibration are proved without it. Under proton lock-in witness discipline [5, 7], laboratory η is a comparison target, not a simultaneous fit row alongside other PDG inputs.

Per the patch-theory reader contract below, the discrete patch layer is the ontology. Smooth-manifold transport, washout, and sphaleron yield are comparison-layer targets—not foundational requirements.

1.5. Dynamic bulk and BBN witnesses (v3). A per-shell matter integrator derives comparison-layer Ω_b and η without importing η_{paper} as an input. Central readout: $\eta_{10} = 6.20$, $\Omega_b = 0.0499$ at lock-in (`data/dynamic_bulk_bbn.v2.json`; §11). At $\eta_{\text{paper}} = 6.10 \times 10^{-10}$, the faithful stoichiometric BBN integrator yields $Y_p = 0.244$, $D/H = 2.51 \times 10^{-5}$, ${}^3\text{He}/H \approx 1.01 \times 10^{-5}$, and ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-depletion; $\approx 45\%$ below Coc semi-analytic 4.65×10^{-10} at the same η [2]; $z(D/H) \approx -0.45$). Legitimate η stages bracket 6.0–6.4 (Tables 2–3). Full abundance certificates remain in [3].

1.6. Proved, witnessed, and deferred.

- **Proved (Lean):** monotonicity of I and Ω_k ; lock-in calibration; parallel η laws; triality tilt averaging; GR/O–Maxwell compatibility; dynamic-bulk ratio packaging ([dynamic bulk vital bundle](#)).

- **Witnessed (Python):** dynamic bulk $\eta_{10} \approx 6.20$, $\Omega_b \approx 0.050$; faithful BBN panel at η_{paper} .
- **Deferred:** associator-derived η_{paper} ; SM washout/sphaleron yield; closed-form η without comparison-layer H_0 , T_{CMB} , or Ω_b .

All numerical tables are regenerated from the standalone Python bundle (§11); no Lean installation is required to verify witnesses.

IMPORTED SPINE (AT A GLANCE)

This note proves ratio-level baryogenesis bookkeeping on the discrete shell ladder. The following inputs are *imported* from earlier Tier-0–2 records; they are listed here so a reader need not open those manuscripts to see what is assumed versus what is derived here. Programme context is in §1.

What this paper adds (new relative to upstream Tier-0–2).

- Horizon-ratio theorems $\Omega_k(n; N) = I(n)/I(N)$ with strict monotonicity and lock-in calibration $\Omega_k(M; M) = 1$.
- Parallel $\eta(n; N)$ law sharing the identical normalisation as Ω_k ([shared-normalisation theorem](#)).
- Triality CP-bias scaffold with proved three-representation averaging ([rapidity CP-bias](#), Appendix A).
- Lean packaging of dynamic-bulk colour-singlet fraction, Ω_b channel, and comparison-layer η bridge ([dynamic bulk vital bundle](#)).
- v3 Python witnesses: per-shell $\eta_{10} \approx 6.20$, $\Omega_b \approx 0.050$, and faithful BBN panel at η_{paper} (Tables 2–3).

TABLE 1. Imported quantities from upstream HQIV papers (not refit here). Lean file paths: Appendix A.

Quantity	Upstream source	Role in this note
$\alpha = 3/5$, $\gamma = 2/5$, shell shape $\sigma(m)$	Tier-0–1: [10, 12, 9]	Curvature imprint ladder; $I(n)$, Ω_k
$\mathcal{N}_{\text{curv}} = 6^7 \sqrt{3}$, $\delta_E(m)$	[10, 8]	Per-shell imprint; octonion/quaternion contrast
$m_{\text{lockin}} = 4$, $\xi_{\text{lock}} = 5$	[7, 5, 10]	Lock-in horizon; $\Omega_k(M; M) = 1$. First shell with full $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ sector spine ($N_{\text{new}}(4) = 40$; octonionic lightcone module, App. A).
$\kappa_6 = \eta_{\text{local}} \gamma C_2(\xi)$, $C_2 = 56/45$ at lock-in	[5]	Matter slot; dynamic- C_2 BBN clock
Lock-in nuclear Q 's (Q_D , Q_{np} , cluster traces)	[4, 5]	Binding-release factor $f_{\text{bind}}(T)$; BBN network
$\eta_{\text{paper}} = 6.1 \times 10^{-10}$ (quarantined)	Paper η anchor module (App. A)	Single absolute anchor; parallel to Ω_k
Comparison H_0 , T_{CMB} , Ω_b	[16, 15] (comparison only)	Late-universe η conversion layer
Observed BBN band (Y_p , D/H ; Coc ${}^7\text{Li}/H \approx 4.65 \times 10^{-10}$)	[2] (comparison only)	Audit targets; HQIV faithful ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ is pre-depletion model output

Derived in this note (not imported). Horizon ratios $\Omega_k(n; N) = I(n)/I(N)$ with strict monotonicity; lock-in calibration $\Omega_k(M; M) = 1$; parallel $\eta(n; N)$ laws sharing the same normalisation; triality CP-bias scaffold; GR/O–Maxwell (α, ϕ) compatibility [9]; dynamic-bulk comparison-layer $\eta_{10} \approx 6.20$ and $\Omega_b \approx 0.050$ without importing η_{paper} (§7, Table 2).

2. SCOPE AND CLAIM DISCIPLINE

The HQIV programme ties discrete null-shell counting, curvature imprint exponent $\alpha = 3/5$ [1], and an eight-dimensional octonionic carrier with machine-checked $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ data [10, 12]. See §1 for programme position and on-ramps.

Lock-in shell $m = 4$ and the G_2 sector spine. The lock-in index $M = 4$ is not an arbitrary fit row: on the discrete null ladder it is the **first shell that carries enough degrees of freedom** to support the full octonion-sector readout used downstream. At $m = 4$ the library proves $N_{\text{new}}(4) = 40$ new lattice modes on the carrier ([octonionic lightcone module](#)), while the algebraic closure $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ supplies fourteen G_2 generators plus the phase-lift harmonic Δ closing to the full 28-dimensional gauge-sector carrier [10]. Proton lock-in, $\Omega_k(M; M) = 1$, and the baryon-asymmetry ratio normalisation are all evaluated at this same row ($\xi_{\text{lock}} = m + 1 = 5$). Earlier shells are pre-sector or pre-closure rungs on the same ladder; later shells extend the chart but do not replace the calibration pin.

Baryogenesis as an application. This note is **not** a standalone theory of baryogenesis; it is one checkpoint on a spine that already shows promising results at every applied scale exercised in the repository so far—hadronic binding, electroweak synthesis, molecular phase geometry, compact-object witnesses, and BBN abundance integration. The formal objective remains the discrete curvature-ratio ledger and its sector projections; η bookkeeping is how that ledger answers a cosmological question at lock-in.

This note answers: **what baryogenesis statement is already a theorem, and what is still interpretive physics?** Geometry and ratios live in the [baryogenesis core module](#) and the [octonionic lightcone module](#); the η_{paper} anchor is isolated in the [paper \$\eta\$ anchor module](#) and wired only through the [baryogenesis witness module](#) (Appendix A).

- (1) **Proved in Lean (formal spine).** Shell indices, ladder temperatures, combinatorial curvature norm $\mathcal{N}_{\text{curv}}$, per-shell imprint $\delta_E(m)$, cumulative integral $I(n)$ with strict monotonicity ([curvature-integral strict-monotonicity theorem](#)), horizon ratios $\Omega_k(n; N)$ with monotonicity, strict monotonicity, and the bound $\Omega_k \leq 1$ for $n \leq N$ ([horizon-ratio monotonicity pack](#)), lock-in calibration $\Omega_k(M; M) = 1$ ([lock-in \$\Omega_k\$ calibration](#)), parallel $\eta(n; N)$ with matching monotonicity lemmas ([\$\eta\$ monotonicity pack](#)), and the identity $\eta = \eta_{\text{paper}} \cdot \Omega_k$ at fixed horizon ([shared-normalisation theorem](#)).
- (2) **Proved dynamic-bulk packaging (v3).** Colour-singlet projector $f_{\text{color}} = 1/3$, baryon channel $(4/8) \cdot (1/3) = 1/6$, dimensionless binding-curvature correction linked to MeV cluster binding, dynamic η at lock-in with dimensionless feedback, and comparison-layer η bridge ([dynamic bulk vital bundle](#); Appendix A).
- (3) **Proved algebraic packaging.** Triality-sensitive increments from the [rapidity CP-bias](#) in the [triality-rapidity-well module](#); affine tilt averaging ([triality tilt average identity](#)); three-representation averaging back to the rapidity well ([triality increment average identity](#)).
- (4) **Structural compatibility (not transport).** The [GR-from-Maxwell module](#): shared ϕ , shared α , and homogeneous HQVM Friedmann equivalence—a compatibility bridge, not a washout-limited yield theorem.
- (5) **Companion: BBN [3].** The BBN formal stack—umbrella certificate, network-from-weights readout, epoch ladder, and integrated witness (Appendix A)—is documented in the companion note; it consumes lock-in η from this note and is *not* part of this note’s proof graph. Coc et al. semi-analytic fits in the library are labelled comparison-only, not HQIV inputs.
- (6) **Yield-level SM physics (not Lean).** Associator/vorticity sources, QCD-window maximisation, and sphaleron rates remain narrative targets beyond both this note and [3]. No continuum-spacetime limit is invoked or required.

3. DISCRETE SHELLS, TEMPERATURES, AND THE BARYOGENESIS WINDOW

Fix the HQIV discrete shell index $m \in \mathbb{N}$ as in [10, 12]. The library defines a QCD onset shell index m_{QCD} and a lock-in shell

$$m_{\text{lockin}} := m_{\text{QCD}} + k_{\text{QCD} \rightarrow \text{lock}},$$

with fixed step count $k_{\text{QCD} \rightarrow \text{lock}}$ (Appendix A), aligned with $M = 4$ (the same lock-in convention used throughout the formalisation and pinned by the proton lock-in witness in [5, 7]). Temperatures on the ladder are $T(m) = 1/(m + 1)$ in natural units (QCD and lock-in temperature closures in the baryogenesis core module).

The finite shell set used for discrete bookkeeping is the explicit **baryogenesis shell window**: a finite interval from m_{QCD} through a few shells after lock-in. This is the honest discrete substitute for a continuum “GeV window”: it is *not* proved from washout maximisation in Lean; it is the packaged combinatorial stage on which curvature ratios are evaluated.

4. CURVATURE IMPRINT, $\mathcal{N}_{\text{curv}}$, AND δ_E

Imported from Tier-0–1 (summary). The shell-shape factor $\sigma(m)$ and curvature exponent $\alpha = 3/5$ enter from the growth trilogy [12, 10]; the octonionic-action note [9] documents the same (α, ϕ) slots in O–Maxwell stationarity. The combinatorial norm $\mathcal{N}_{\text{curv}}$ and per-shell imprint $\delta_E(m)$ are proved in the lightcone module—not re-derived here. See Table 1.

Let $\sigma(m)$ denote the shell-shape factor (the **shell-shape factor**), built from the same α -weighted logarithmic curvature density used in the growth trilogy. The combinatorial curvature norm is the exact real scalar $\mathcal{N}_{\text{curv}} = 6^7 \sqrt{3}$, proved in the lightcone module from the structural “six cube directions, seven imaginary octonion directions, half-diagonal $\sqrt{3}$ ” bookkeeping (the **curvature-norm structural theorem** and related lemmas). The per-shell imprint is $\delta_E(m) = \mathcal{N}_{\text{curv}} \cdot \sigma(m)$ (the **QCD-shell imprint** specialises at m_{QCD}).

Structural rigidity (octonionic vs. quaternionic comparison). Lean also proves that a natural quaternionic Maxwell comparison norm undershoots the HQIV octonionic norm by a rigid factor 1296 shellwise (the **octonion–quaternion imprint ratio theorem**). This supports the narrative that $\mathcal{N}_{\text{curv}}$ is not an arbitrary normalisation: it is the imprint constant selected by the octonionic lift inside the same eight-dimensional carrier used for gauge closure.

5. HORIZON RATIOS Ω_k AND LOCK-IN CALIBRATION

Define the cumulative partial integral

$$I(n) := \sum_{m=0}^{n-1} |\sigma(m)|$$

(the **curvature integral**). For horizons N with $I(N) > 0$, the library defines $\Omega_k(n; N) := I(n)/I(N)$. At the lock-in shell, Lean proves positivity $I(m_{\text{lockin}}) > 0$ and the calibration $\Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1$ (the **lock-in Ω_k calibration**). This is the precise sense in which “spatial curvature lock-in” is a **theorem**: normalisation of the discrete imprint integral at the declared reference horizon, not an experimental input.

Proposition 5.1 (Monotonicity and approach to lock-in, Lean). *Assume $I(N) > 0$. Then for all $n_1 \leq n_2$, $\Omega_k(n_1; N) \leq \Omega_k(n_2; N)$, with strict inequality when $n_1 < n_2$ (the **horizon-ratio monotonicity pack**). If additionally $n \leq N$, then $\Omega_k(n; N) \leq 1$. In particular, along shells $n \leq m_{\text{lockin}}$ with horizon $N = m_{\text{lockin}}$, the ratio $\Omega_k(n; m_{\text{lockin}})$ increases strictly toward 1 as n approaches m_{lockin} from below.*

6. BARYON ASYMMETRY η AS THE SAME NORMALIZATION (OPTIONAL ABSOLUTE ANCHOR)

Without importing any experimental η . Everything in the baryogenesis core module is logically independent of PDG-scale baryon asymmetry.

With a single absolute anchor (quarantined). The constant η_{paper} appears *only* in the [baryogenesis \$\eta\$ paper module](#). The witness module defines $\eta(n; N) := \eta_{\text{paper}} \cdot I(n)/I(N)$ when $I(N) > 0$ (and a conservative fallback otherwise), and proves the parallel identities to the Ω_k ratio (the [horizon \$\eta\$ readout](#) and [self-horizon \$\eta\$ identity](#) at $n = N$).

The conceptual punchline is already a theorem:

$$\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$$

(the [shared-normalisation theorem](#)). At fixed horizon, baryogenesis is not a second independent imprint law: it is the same partial-integral ratio as Ω_k , scaled by one overall anchor.

Same monotonicity and bounds as Ω_k . Lean also proves $\eta(n_1; N) \leq \eta(n_2; N)$ for $n_1 \leq n_2$ (the [\$\eta\$ monotonicity pack](#)), and $\eta(n; N) \leq \eta_{\text{paper}}$ whenever $n \leq N$. Thus η_{paper} is both the absolute scale at lock-in and an upper envelope for readouts inside the horizon—a direct consequence of shared monotonicity with Ω_k .

Honesty boundary (first principles vs. calibration). Lean does not currently derive η_{paper} from associator phases or SM chemical potentials. What *is* first-principles inside the formalisation is the shared ratio law and the lock-in calibration of Ω_k . Attaching the observed η scale is a *single-point calibration* quarantined to one module—analogue to how the mass and Kirchhoff papers treat amplitude normalisation while keeping imprint *shape* combinatorially rigid.

7. DYNAMIC BULK MATTER AND METHOD RANGE FOR η

The ratio spine of §2–§5 fixes how $\eta(n; N)$ is *normalized* once an absolute anchor is declared. A separate Python witness chain now asks what η is when that anchor is *not* imported: baryon budget is accumulated shell-by-shell from curvature imprint, Casimir inside/outside balance, and composite-trace binding feedback, then converted to η only at a late comparison layer using present-day H_0 , T_{CMB} , and Ω_b .

Per-shell integrator (v3). The dynamic bulk witness (§11) evolves shells $m_{\text{QCD}} \leq m \leq m_{\text{lock}}$ with emergent hadronic lock-in at $m = 4$ ($\xi_{\text{lock}} = 5$). At each step it reads shell shape $\sigma(m)$, inner/outer curvature proxies, the per-shell curvature budget $B_{\text{curv}}(m)$ (early matter–antimatter / radiation seed relaxing to 1 at lock-in; excess imprint outside the baryon track), a temperature-relative vev factor from the inner/outer Casimir balance (described below), and dynamic proton mass and nuclear Q values from the 8×8 composite-trace network. Matter increments

$$\Delta S_m = \sigma(m) f_{\text{vev}}(\xi_m) + w_{\text{strong}} f_{\text{bind}}(m, \xi_m) B_{\text{curv}}(\xi_m)$$

accumulate into a cumulative source $C = \sum_m \Delta S_m$. Baryon and total matter fractions follow

$$\Omega_m = \gamma C / (C + \rho_{\text{rad}}), \quad \Omega_b = \Omega_m \cdot w_{\text{strong}} \cdot f_{\text{color}},$$

with HQIV structural weights $\gamma = 2/5$, $w_{\text{strong}} = 4/8$, and a colour-singlet projector $f_{\text{color}} = 1/3$ (trace-derived sweep below). The inner/outer Casimir scale uses a dynamic *outer* suppression $\omega K(\xi)/140$ (not a fixed $1/140$ at all epochs; the integer 140 counts discrete fluctuation modes on the outer neutral horizon shell at lock-in, while the amplitude runs with $\omega K(\xi)$ like the inner trapped-Casimir factor). The witness reports $C = 3.80$, $\Omega_b = 0.0499$, $\Omega_m = 0.299$ at $m = 4$ (`dynamic_bulk.bbn_v2.json`, lock-in row).

What “inside” and “outside” mean (reader-facing). HQIV treats the octonionic carrier as having two competing Casimir-like surfaces on the *same* geometric object, not two unrelated constants.

- **Inside (contact shells).** On the three inner Hopf contact shells (the heavy-sector winding ladder), modes are “trapped” by phase-lift and Beltrami torsion. This inner surface sets the

scale of binding, the heavy-lepton gap, and the hadronic chart. Its strength grows with the curvature readout $\omega K(\xi)$ on the temperature ladder.

- **Outside (neutral horizon shell).** On the first shell *beyond* proton lock-in, fluctuations on the right-handed neutrino/neutral channel of the same carrier are coarse-grained into a finite number of discrete modes. At lock-in that count is 140, giving the familiar factor $1/140$ when the curvature amplitude is normalized to unity. Away from lock-in the outer suppression is $\omega K(\xi)/140$, parallel to the inner side.

The overall mass scale and vev factor are the ratio inner/outer. The formal library records this as the *outer neutral-shell fluctuation witness* (Appendix A)—not a standard physics abbreviation and not a particle label.

Comparison-layer η . Only after the shell ladder is integrated does the witness form

$$\eta = \frac{n_b}{n_\gamma} = \frac{\Omega_b \rho_{\text{crit}}}{m_p n_\gamma(T_{\text{CMB}})},$$

with default $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $T_{\text{CMB}} = 2.7255 \text{ K}$ [16, 15]. This is **not** part of the proved Lean ratio graph; it is an auditable late-universe comparison readout. Lean modules [Dynamic BBN and baryogenesis](#) and [shell reaction opportunity](#) now mirror the binding-release and shell-clock factors used in the witness.

Why η_{10} moved: $6.09 \rightarrow 6.36 \rightarrow 5.59$. The three numbers are not random drift; they label different stages of the programme:

- **6.10 (legacy anchor).** η_{paper} in the quarantined [baryogenesis \$\eta\$ paper module](#); still the lock-in calibration used by the fixed- η BBN witness.
- ≈ 6.28 – 6.36 (**intermediate dynamic paths**). Early dynamic-baryogenesis witness with horizon dynamic η and binding-curvature correction ($\eta_{10} = 6.28$; Appendix A); alternatively, holding the dynamic bulk Ω_b fixed and raising only the comparison H_0 to 70 – $72 \text{ km s}^{-1} \text{ Mpc}^{-1}$ gives $\eta_{10} = 6.04$ – 6.39 .
- **6.20 (current bulk v3 central value).** Full per-shell Ω_b accumulation with dynamic vev, B_{curv} on shells, and binding feedback, then comparison-layer conversion at $H_0 = 67.4$.

None of these replaces the proved identity $\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$; they test how much of the observed scale can be *derived* before the single-point anchor is applied.

Fixed η_{paper} vs. dynamic bulk (one sentence). All ratio-level theorems (monotonicity, lock-in calibration, shared normalisation) are independent of the numerical value of η_{paper} . The dynamic-bulk integrator tests how much of the observed scale can be recovered from per-shell curvature, Casimir vev, and binding feedback *before* the single-point anchor is applied.

TABLE 2. Comparison-layer $\eta_{10} \equiv \eta \times 10^{10}$ from different HQIV stages and inputs. Errors on the dynamic-bulk central value propagate $H_0 = 67.4 \pm 2.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at fixed Ω_b ; other rows are deterministic readouts at stated inputs. Observed band from BBN+*CMB* synthesis [2].

Method	η_{10}	uncertainty	Notes
Observed (BBN+ <i>CMB</i>)	6.10	± 0.06	Comparison target [2]
Lean lock-in anchor (η_{paper})	6.10	—	Quarantined calibration; not derived here
Early dynamic baryogenesis	6.28	—	Horizon dynamic η + binding correction
Dynamic bulk v3 (primary)	6.20	± 0.42	Per-shell Ω_b ; B_{curv} + dynamic C_2 BBN clock
$H_0 = 70$ at fixed Ω_b	6.04	—	Comparison-layer H_0 sensitivity
$H_0 = 72$ at fixed Ω_b	6.39	—	Upper bracket of H_0 sweep
$\rho_{\text{rad}} = 0.5$ vs. 1.0	6.68 / 5.60	—	Radiation-floor sensitivity on shell source

Casimir-modulated binding and BBN clock. Bound-state Casimir trapping in the early curvature background changes effective nuclear Q values during BBN. The witness uses a curvature–temperature binding-release factor

$$f_{\text{bind}}(T) = \exp\left(-\gamma w_{\text{strong}} \frac{s}{1+s}\right), \quad s = \frac{\Omega_k(\xi) - \Omega_k(\xi_{\text{lock}})}{\log(\xi/\xi_{\text{lock}})},$$

applied to lock-in nuclear cluster Q ’s (inside/outside composite trace and caustics; [4]) rather than raw $\xi = T_{\text{pl}}/T$ vev blow-up. For BBN network integration the native clock is a **shell reaction opportunity**

$$\mathcal{O}_{\text{shell}} = \Delta \log \xi \cdot [\log(\xi/\xi_{\text{lock}})]^3 B_{\text{curv}}(\xi(T)) f_{C_2}(T) (1 + \delta_{\text{bind}}/4),$$

where B_{curv} is the same-epoch Casimir/nuclear outside budget in the κ_6 matter slot ([5]), and

$$T_{\text{bn}}(\eta) = \gamma_{\text{HQIV}} w_{\text{strong}} T_{\text{freeze}}(\eta), \quad T_{\text{freeze}}(\eta) = \frac{Q_{np}}{\log(\eta_{10})}, \quad w_{C_2}(T) = \gamma_{\text{HQIV}} w_{\text{strong}} \frac{Q_D^{\text{eff}}(T)}{Q_{np}},$$

$$f_{C_2}(T, \eta) = \begin{cases} (\kappa_{6,\text{ref}}/\kappa_6(\xi(T)))^{w_{C_2}}, & T \leq T_{\text{bn}}(\eta), \\ 1, & T > T_{\text{bn}}(\eta), \end{cases} \quad \kappa_6 = \eta_{\text{paper}} B_{\text{curv}} \gamma C_2(\xi),$$

with $\kappa_{6,\text{ref}}$ at $T_{\text{ref}} = T_{\text{freeze}}(\eta)$, $w_{\text{strong}} = 4/8$, and $Q_D^{\text{eff}}(T)$ from the effective deuteron binding readout at temperature (Appendix A). No fitted MeV cutoffs: $w_{C_2}(T)$ rises toward ≈ 0.36 in the bottleneck and $T_{\text{bn}} \approx 0.14$ MeV at $\eta_{10} \approx 6.2$ are *derived*. Raw κ_6 growth on the hot chart over-burns D; the inverse lapse clock is the BBN readout of dynamic C_2 . Standard radiation-dominated $H(T)$ is retained only as a diagnostic comparison (legacy H -cooling mode).

BBN witnesses vs. standard reaction networks. The custom shell-opportunity and dynamic- C_2 mechanisms are Lean-aligned witnesses that modulate the deuterium bottleneck and effective Q -values; they are **not** claimed to be equivalent to the full standard BBN reaction network. Full stoichiometric integration results (Table 3, [3]) are reported for comparison only; the proved content of this note stops at ratio-level η bookkeeping and the dynamic-bulk comparison layer.

TABLE 3. BBN abundance readout at lock-in $\eta_{\text{paper}} = 6.10 \times 10^{-10}$ (faithful integrator; §11). Shell-opportunity scaffold row documents the pre-synthesis-window driver; full abundance theorems are in [3].

Network driver	Y_p	D/H	${}^7\text{Li}/H$
Faithful stoichiometric integrator (primary)	0.244	2.51×10^{-5}	2.54×10^{-10}
Shell opportunity + dynamic C_2 (scaffold)	—	$\ll 10^{-5}$	—
Legacy $H(T)$ + fixed lock-in Q	0.249	5.3×10^{-10}	—
Coc semi-analytic ([2])	0.244	2.58×10^{-5}	4.65×10^{-10}
Spite-plateau band	—	—	$(1.6\text{--}4.5) \times 10^{-10}$
Observed ($Y_p, D/H$)	0.244 ± 0.004	$(2.53 \pm 0.04) \times 10^{-5}$	—

Row notes. Primary row: ${}^3\text{He}/H \approx 1.01 \times 10^{-5}$; pre-depletion ${}^7\text{Li}$; $z(D/H) \approx -0.45$ vs. Coc [2]. Coc and Spite rows are comparison layers, not HQIV integrator inputs.

Honest error budget. The dynamic bulk central value sits $\approx 8\%$ below the observed η_{10} at default H_0 . Propagating only the comparison-layer H_0 uncertainty ($\pm 2.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$) gives $\eta_{10} = 5.60 \pm 0.42$. A parallel early-dynamic baryogenesis correction and an H_0 -only raise bracket 6.0–6.4 without changing the per-shell source. Remaining upward levers visible in the witness—not yet promoted to theorems—include a trace-derived colour-singlet projector above 1/3 and the binding-sourced baryogenesis correction in the horizon dynamic η readout (Appendix A). Downward levers include a

larger effective radiation floor in the shell partition and stopping the integrator before additional post-lock-in shell accumulation.

Witness discipline. Machine-checked JSON witnesses are listed in §11. Standalone reproduction bundle: `scripts.zip` in this folder.

8. CP BIAS WITHOUT NAMING η : TRIALITY COUPLING

The [triality-rapidity-well module](#) defines

$$\text{CP-bias}(m) := \Omega_k(m; m_{\text{lockin}}) - 1,$$

the [rapidity CP-bias](#)—a pure curvature-ratio deviation at the lock-in reference horizon. It then builds representation-sensitive triality increments by multiplying baseline barrier surfaces by $(1 + \text{CP-bias}(m) \cdot w_{\text{rep}})$ with explicit $w_{\text{rep}} \in \{0, +1, -1\}$ on the three $\text{Spin}(8)$ triality channels, and proves exact three-rep averaging back to the rapidity well.

Auditable zero-sum tilt. Writing the weights as $w_{8_v} = 0$, $w_{8_{s+}} = +1$, $w_{8_{s-}} = -1$, Lean proves the affine factors average identically to unity:

$$(1) \quad \frac{1}{3} \sum_{\text{rep} \in \{8_v, 8_{s+}, 8_{s-}\}} (1 + w_{\text{rep}} \cdot \text{CP-bias}(m)) = 1$$

([triality tilt average identity](#)). The full triality increments average to the rapidity-written well surface ([the triality increment average identity](#)).

This is the cleanest Lean-backed “CP-odd bias is tied to the same curvature-ratio channel” statement that does *not* import η_{paper} . It is still **not** a replacement for CPT/sphaleron/washout dynamics; it is a structured, finite-dimensional coupling scaffold consistent with the lock-in ratios.

9. LINK TO HQVM FRIEDMANN AND O-MAXWELL (COMPATIBILITY, NOT YIELD)

The [GR-from-Maxwell module](#) packages that the same auxiliary ϕ appears in the HQVM lapse story and the O-Maxwell bookkeeping (the [shared- \$\phi\$ identity](#)), and that the same lattice $\alpha = 3/5$ enters $G_{\text{eff}}(\phi) = \phi^\alpha$ on the Friedmann side (the [shared- \$\alpha\$ identity](#)). The homogeneous Friedmann identity is packaged as an explicit algebraic equivalence (the [HQVM Friedmann compatibility theorem](#)).

This supports the *narrative* that cosmological background evolution used in companion numerics is not stitched from unrelated parameters: it is the same (α, ϕ) ledger as the discrete gauge sector [9]. It does *not*, by itself, integrate Boltzmann equations for baryon number violation in the SM.

10. WHAT WOULD COMPLETE A “FULL DYNAMICAL DERIVATION”

To go beyond ratio bookkeeping and triality scaffolds, a future formalisation would need explicit discrete transport theorems on the same shell indices, for example:

- a well-posed chemical-potential / Boltzmann layer tied to shell indices;
- a washout bound as a theorem about monotone functionals of $I(n)$ under a declared interaction kernel;
- a provable map from non-associative phase holonomy to a signed baryon production density, with finite totals controlled by $I(n)$.

None of these are discharged in the current repository. Smooth-manifold rewrites of such transport are comparison-layer targets only—not a continuum-completion programme. The present manuscript is the maximal honest baryogenesis paper aligned to actual Lean artifacts.

COMPANION: BIG-BANG NUCLEOSYNTHESIS [3]

This note stops where BBN begins at the *ratio-level* lock-in calibration $\Omega_k(M; M) = 1$ and the shared $\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$ readout. The companion BBN preprint documents the library stack: the [BBN umbrella module](#) re-exports a network readout from composite-trace binding and isotope-ladder weights ([BBN network-from-weights module](#)), an epoch ladder ([BBN epoch modules](#)), and an integrated witness certificate ([BBN witness module](#)) checked against the BBN witness export (§11). Standard semi-analytic Y_p and D/H reference fits (Coc et al.) appear only as *comparison layers*, not as HQIV-derived inputs.

The present v3 update adds the dynamic-bulk path that *feeds* comparison-layer η and documents faithful BBN readouts at lock-in η_{paper} : per-shell Ω_b from the dynamic bulk witness, Casimir-modulated binding Q 's [4], and the stoichiometric integrator at η_{paper} (Tables 2–3; §11). Those witnesses determine the comparison-layer η band and the lock-in abundance panel passed downstream; full abundance theorems remain in [3].

11. CONCLUSION

The first-principles content formalised today is:

- (1) a rigid combinatorial imprint norm $\mathcal{N}_{\text{curv}} = 6^7 \sqrt{3}$ with an octonionic/quaternionic contrast theorem;
- (2) a discrete cumulative curvature integral $I(n)$ and horizon ratios $\Omega_k(n; N)$ with proved strict monotonicity of I , monotonicity (and strict monotonicity) of Ω_k in the readout shell, the bound $\Omega_k(n; N) \leq 1$ for $n \leq N$, and lock-in calibration $\Omega_k = 1$ at m_{lockin} ;
- (3) a parallel $\eta(n; N)$ law sharing the *identical* normalization as $\Omega_k(n; N)$, with matching monotonicity lemmas and $\eta(n; N) \leq \eta_{\text{paper}}$ for $n \leq N$, and with η_{paper} quarantined to a single module;
- (4) a triality representation scaffold coupling curvature-ratio bias to CP-sensitive increments, without importing η_{paper} . This scaffold ties CP-odd increments to the same curvature-ratio channel used for Spin(8) closure and measurement; it does *not* derive the bias itself from non-associativity of the octonion product (that remains a Tier-IV target).
- (5) a GR/O–Maxwell compatibility package tying (α, ϕ) across sectors;
- (6) (**v3 witnesses**) a dynamic bulk integrator deriving comparison-layer $\eta_{10} \approx 6.20$ and $\Omega_b \approx 0.050$ from per-shell curvature, Casimir vev, and B_{curv} ; at lock-in η_{paper} the faithful BBN integrator yields $Y_p \approx 0.244$, $D/H \approx 2.51 \times 10^{-5}$ ($z \approx -0.5$ vs. Coc [2]), ${}^7\text{Li}/H \approx 2.54 \times 10^{-10}$ (pre-depletion; Coc comparison $\approx 4.65 \times 10^{-10}$; Tables 2–3).

That package is already enough for a mathematically disciplined “baryogenesis stands on the same shoulders as Spin(8) closure” paper, now augmented by an explicit method-range audit for the observed η band.

Forward path. The natural next formal manuscript is the BBN stack [3]. The remaining distance to textbook electroweak baryogenesis lies in washout, sphaleron integration, and associator-derived CP violation—yield-level dynamics not yet in the Lean corpus. Readers interested in the full programme are invited to begin with [disregardfiat.tech](#) and the [HQIV-LEAN repository](#). Closing the $\approx 8\%$ gap between the dynamic-bulk central η_{10} and the observed value is expected to come from trace-derived baryon projectors and binding-sourced baryogenesis corrections already scaffolded in Lean, not from refitting η_{paper} .

REPRODUCIBILITY

All numerical rows in Tables 2–3 are regenerated from standalone Python witnesses bundled in `scripts.zip` in this folder (build script: [bundle_baryogenesis_scripts.py](#) from the repository root). The bundle includes entry scripts, transitive `hqiv_*` imports, mirrored witness JSON,

`pyproject.toml`, and `MANIFEST.sha256`. No Lean installation is required to verify the witnesses. The same scripts power witness panels on disregardfiat.tech.

Repository. Source of truth: github.com/HQIV/hqiv-lean. Witness JSON at `data/` in the repository root; the deposit bundle mirrors these under `scripts/data/`.

Quick regeneration (repository root).

```
python3 scripts/hqiv_dynamic_bulk_bbn.py
python3 scripts/hqiv_bbn_integrator.py --json data/bbn_integrator.json
python3 scripts/hqiv_integrator_lean_audit.py
python3 -m unittest scripts.test_hqiv_dynamic_c2_bbn \
    scripts.test_hqiv_bbn_integrator
python3 scripts/bundle_baryogenesis_scripts.py
From the extracted bundle only (no Lean build required):
cd papers/baryogenesis_lockin/scripts
PYTHONPATH=. python3 hqiv_dynamic_bulk_bbn.py
PYTHONPATH=. python3 hqiv_bbn_integrator.py --json data/bbn_integrator.json
PYTHONPATH=. python3 hqiv_integrator_lean_audit.py
```

ACKNOWLEDGMENTS

AI-assisted drafting and repository tooling were used during manuscript preparation. All formal claims were checked against the cited Lean modules in the HQIV-LEAN library [14].

APPENDIX A. LEAN MODULE INDEX

The Lean development lives at [HQIV-LEAN](https://github.com/HQIV-LEAN) [14]. The main text cites *English* module and theorem names (clickable links to entries below); Lean file paths and identifier names appear only in this appendix. The same index is browsable at disregardfiat.tech.

Recommended starting modules for newcomers.

[Hqiv/Geometry/OctonionicLightCone.lean](#):

Discharges the discrete shell ladder, shell-shape factor $\sigma(m)$, curvature integral $I(n)$, and Ω_k monotonicity—the geometric spine every downstream application shares.

[Hqiv/Physics/BaryogenesisCore.lean](#):

Packages QCD/lock-in shell indices and Ω_k readouts *without* importing experimental η ; start here for ratio-level cosmology.

[Hqiv/Physics/BaryogenesisWitness.lean](#):

Defines $\eta(n; N)$ with shared normalisation to Ω_k and proves the parallel monotonicity lemmas once η_{paper} is declared.

[Hqiv/Physics/BaryogenesisDynamicBulk.lean](#):

v3 dynamic-bulk packaging: colour-singlet fraction, Ω_b channel, comparison-layer η bridge (witnessed in Python, partially proved).

[Hqiv/Physics/BBNStoichiometricIntegrator.lean](#):

Companion entry to [3]: faithful BBN stoichiometric integrator consuming lock-in η from this note.

Full module index.

Octonionic lightcone module.:

Shell shape, curvature integral, Ω_k monotonicity pack, combinatorial curvature norm. [Hqiv/Geometry/OctonionicLightCone.lean](#).

Shell-shape factor.:

`shell_shape`.

Curvature integral.:

$$I(n) = \sum_{m < n} |\sigma(m)|.$$
code: `curvature_integral.`
Curvature-integral strict monotonicity.:code: `curvature_integral_strict_mono.`**Curvature-norm structural theorem.:**code: `curvature_norm_determined_by_structure.`**QCD-shell imprint.:**code: `deltaE_at_QCD_shell.`**Octonion-quaternion imprint ratio.:**code: `deltaE_eq_1296_mul_deltaE_quaternionicCandidate.`**Baryogenesis core module.:**Shell indices, temperatures, Ω_k without η . [Hqiv/Physics/BaryogenesisCore.lean](#).**QCD and lock-in temperature closures.:**code: `T_QCD_closed, T_lockin_closed.`**Baryogenesis shell window.:**code: `baryogenesisShells.`**Horizon-ratio monotonicity pack.:**code: `omega_k_at_horizon_mono, omega_k_at_horizon_strict_mono, omega_k_at_horizon_le_one_of_le.`**Lock-in Ω_k calibration.:**code: `omega_k_lockin_calibration.`**Baryogenesis η paper module.:** η_{paper} only. [Hqiv/Physics/BaryogenesisEtaPaper.lean](#).**Baryogenesis witness module.:** η monotonicity and shared normalisation. [Hqiv/Physics/BaryogenesisWitness.lean](#).**Horizon η readout.:**code: `eta_at_horizon_eq.`**Self-horizon η identity.:**code: `eta_at_horizon_self.` **η monotonicity pack.:**code: `eta_at_horizon_mono, eta_at_horizon_strict_mono, eta_at_horizon_le_eta_paper.`**Shared-normalisation theorem.:**code: `baryogenesis_same_normalization_as_omega_k.`**Triality-rapidity-well module.:**[Hqiv/Physics/TrialityRapidityWellEquivalence.lean](#).**Rapidity CP-bias.:**code: `rapidityCPBias.`**Triality tilt average identity.:**code: `triality_cp_tilt_factors_average_eq_one.`**Triality increment average identity.:**code: `cpSensitiveTrialityIncrement_threeRep_average_eq_rapidityWell.`**GR-from-Maxwell module.:**[Hqiv/Physics/GRFromMaxwell.lean](#).**Shared- ϕ identity.:**code: `same_phi_in_0_Maxwell_and_HQVM.`**Shared- α identity.:**code: `same_alpha_in_0_Maxwell_and_HQVM.`**HQVM Friedmann compatibility.:**code: `O_Maxwell_compatible_with_HQVM_GR_homogeneous.`

BBN stoichiometric integrator (companion):

Synthesis-window D/H , coupled neutron inventory, stratified ^3He gates; Python mirror [scripts/hqiv_bbn_integrator.py](#). [Hqiv/Physics/BBNStoichiometricIntegrator.lean](#).

BBN companion [3] (not in this note's graph):

Umbrella + network + epoch + witness; consumes lock-in η . See entries below.

BBN umbrella module:

Combined vital readout certificate. [Hqiv/Physics/BigBangNucleosynthesis.lean](#).

BBN network-from-weights module:

Primary light-element readout from HQIV weights. [Hqiv/Physics/BBNNetworkFromWeights.lean](#).

BBN epoch modules:

Epoch ladder and evolution. [Hqiv/Physics/BBNEpochNetwork.lean](#), [Hqiv/Physics/BBNEpochEvolution.lean](#).

BBN witness module:

Integrated witness + Python data/bbn_witnesses.json. [Hqiv/Physics/BBNWitness.lean](#).

Dynamic BBN and baryogenesis module:

Curvature-temperature binding release, effective BBN Q 's at temperature, shell reaction opportunity with binding feedback. [Hqiv/Physics/DynamicBBNBaryogenesis.lean](#).

BBN dynamic- C_2 pack:

`bbnKappa6AtT_MeV`, `bbnLapseConcentrationAtT_MeV`, `bbnDynamicC2BottleneckT_MeV`, `bbnDynamicC2LapseExponent`, `bbnDynamicC2OpportunitySuppression`, `bbnShellReactionOpportunity_dynamic_integrator` in [Hqiv/Physics/DynamicBBNBaryogenesis.lean](#); `tuftHopfKappa6AtXi`, `tuftLapseConcentrationAtXi` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

Shell reaction opportunity:

`bbnShellReactionOpportunity`, `bbnCurvatureOpportunityFactor` in [Hqiv/Physics/BBNEpochNetwork.lean](#).

Dynamic bulk vital bundle:

`colorSingletFraction`, `omegaBFromOmegaM`, `etaComparisonFromCosmology`, `baryogenesis_dynamic_bulk_vital_holds_any`. [Hqiv/Physics/BaryogenesisDynamicBulk.lean](#).

Dynamic bulk witness (Python):

`scripts/hqiv_dynamic_bulk_bbn.py` $\rightarrow$ `data/dynamic_bulk_bbn_v2.json`;
`scripts/hqiv_integrator_lean_audit.py` $\rightarrow$ `data/integrator_lean_audit.json`.

APPENDIX B. LEAN DEPENDENCY SKETCH

OctonionicLightCone

```
(shell_shape, curvature_integral, curvature_integral_strict_mono,
 omega_k_at_horizon_mono / _strict_mono / _le_one_of_le,
 curvature_norm_combinatorial, deltaE, referenceM, qcdShell, ...)
|
v
```

BaryogenesisCore

```
(m_QCD, m_lockin, T_QCD, T_lockin, baryogenesisShells,
 omega_k_lockin_calibration, ...)
|
+-----> TrialityRapidityWellEquivalence
|          (rapidityCPBias; triality tilt average)
|
+--> BaryogenesisEtaPaper (eta_paper only)
|
v
```

BaryogenesisWitness

```
(eta_monotonicity; baryogenesis_same_normalization_as_omega_k)
```

DynamicBBNBaryogenesis + HopfShellBeltramiMassBridge (dynamic C2 pack)
 GRFromMaxwell (same_phi / same_alpha / Friedmann compatibility)

BigBangNucleosynthesis (companion; not in this note's graph)
 --> BBNNetworkFromWeights, BBNEpochNetwork, BBNWitness

Python witnesses: see Sec. repro

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